Statistical Modeling Support Group

PART B: ESSAY PRACTICE PAPER

20 extended questions | 25 practice marks per question

Preparation approach. Questions 1–7 emphasize inference and regression; 8–13 cover classification; 14–16 cover response distributions and Bayesian methods; 17–20 cover experimental design, PCA and time series. Use the appendix only after attempting a question.

Question 1 [25 marks]

Comparing independent population means

A researcher compares cholesterol concentrations in independent random samples from two populations, A and B. Assume both populations are normal, with unknown variances that are not assumed equal.

A	195	225	265	231	225	236	245	235	241	226
B	224	192	197	211	208	235	229	231	228	219

The claim is that the population mean difference $\mu_A - \mu_B$ differs from 10 units. Use a two-sided test at 5%.

(a) Define the parameter and state both hypotheses. Identify the appropriate test and explain why an independent-samples procedure is required. [5 marks]

(b) Calculate the sample means, sample variances, standard error and test statistic. Give the approximate reference degrees of freedom. [5 marks]

(c) Obtain or specify the calculation of the p-value, make a decision, and write a conclusion addressing the claimed difference of 10 units. [5 marks]

(d) Construct a two-sided 95% confidence interval for $\mu_A - \mu_B$. Explain its relationship with the hypothesis-test conclusion. [5 marks]

(e) Write R statements to enter the observations, perform the required test and obtain the interval. Explain how the procedure changes if equal population variances are justified. [5 marks]

Question 2 [25 marks]

Paired measurements and the direction of a claim

Eight students complete comparable tests before and after a workshop. Each column is a matched student pair. Different students are independent. Assume the differences are normally distributed.

Before	52	60	55	48	70	65	58	62
After	57	63	59	50	76	69	63	65

The department wants evidence that the mean improvement exceeds 2 marks.

(a) Define the paired difference and formulate the null and directional alternative. Explain why treating all 16 scores as independent is inappropriate. [5 marks]

(b) Calculate the mean and standard deviation of the differences, the standard error and the test statistic. State its null reference distribution. [5 marks]

(c) Calculate the one-sided p-value and state the practical conclusion at 5%. Also calculate a two-sided 95% interval for the mean improvement. [5 marks]

(d) Write R code that creates the paired data, performs the directional test and checks the normality condition relevant to this design. [5 marks]

(e) Distinguish the conclusions “the scores improved and “the workshop caused the improvement. Propose a randomized comparison that would strengthen causal interpretation. [5 marks]

Question 3 [25 marks]

Proportions, confidence intervals and variance inference

Process A produces 24 defective components among 200 sampled; independent process B produces 45 among 250. Samples are random and the normal approximation for proportions is adequate. Separately, 20 independent measurements from a normal manufacturing population have sample variance $s^2 = 9$ (units squared), with an unknown mean. The claimed population variance is 4.

(a) Test whether process A has a lower defect probability than process B. State the hypotheses, pooled proportion, standard error, signed statistic and one-sided p-value. [5 marks]

(b) Calculate a two-sided 95% Wald interval for $p_A - p_B$ using an unpooled standard error. Explain why this interval can contain zero while the one-sided test in (a) is significant. [5 marks]

(c) Write R commands for the proportion test without continuity correction and for the interval in (b). Explain what quantity the test statistic reported by `prop.test` represents. [5 marks]

(d) For the separate normal sample, test whether the population variance exceeds 4. Calculate the statistic, degrees, upper-tail p-value and decision. [5 marks]

(e) Write R code for the variance test and a two-sided 95% interval for the variance. State the normality assumption and explain why substituting standard deviations into a variance formula would be wrong. [5 marks]

Question 4 [25 marks]

Completely randomized design, ANOVA and Fisher LSD

Twelve comparable plots are randomly assigned to three fertilizers, four plots per treatment. Yield measurements are shown below. Assume independent normal errors and equal error variances.

Fertilizer A	18	20	19	21
Fertilizer B	22	24	23	25
Fertilizer C	26	28	27	29

For hand calculations use $t_{0.975,9} = 2.2622$ and $F_{0.95;2,9} = 4.2565$, where subscripts specify lower-tail quantiles.

(a) Identify the experimental unit, factor, treatments, response and number of independent replications per treatment. State the CRD model and assumptions. [5 marks]

(b) State the ANOVA hypotheses. Calculate treatment means, grand mean and the treatment, error and total sums of squares. [5 marks]

(c) Complete the ANOVA table with degrees of freedom, mean squares and the treatment F statistic. Make a 5% decision and interpret it. [5 marks]

(d) Calculate the common Fisher LSD for a pair of treatment means and identify which pairs differ. Explain what the omnibus ANOVA did not tell you and state a multiple-comparison limitation of unadjusted LSD. [5 marks]

(e) Write R code to create the data, fit the ANOVA, inspect residual normality and equal variances, and perform Fisher LSD. Explain why a large Levene-test p-value is not proof of equal variances. [5 marks]

Question 5 [25 marks]

Regression in matrix form and multiple responses

Five independent units have predictor x and two continuous responses y_1, y_2:

x	0	1	2	3	4
y_1	2	3	7	7	11
y_2	1	3	2	5	4

Fit an intercept and a slope for each response using least squares. The inverse cross-product matrix is $(X^\top X)^{-1} = \begin{pmatrix} 0.6 & -0.2 \\ -0.2 & 0.1 \end{pmatrix}$.

(a) Write the design matrix X, the response matrix Y and their dimensions. Distinguish multivariate regression here from a model with several predictors and one response. [5 marks]

(b) Calculate $X^\top Y$ and the estimated coefficient matrix $\widehat{B} = (X^\top X)^{-1} X^\top Y$. State both fitted equations. [5 marks]

(c) Calculate the fitted-value and residual vectors for y_1, its residual sum of squares and its ordinary R^2. Interpret R^2 without making a causal claim. [5 marks]

(d) Write R code that creates the data, fits each response, obtains a Pearson correlation test between x and y_1, and predicts both responses at $x = 2.5$. [5 marks]

(e) Now suppose a four-level nominal region predictor is added. Explain the indicator coding and reference category, the new design-matrix dimensions, and why using all four indicators with an intercept creates a problem. [5 marks]

Question 6 [25 marks]

Diagnostics and selection of a multiple regression model

A continuous outcome is modelled using 60 independent units and six numerical candidate predictors. The following are deliberately supplied practice results, not output that must be regenerated from an unavailable file.

Predictor count	Training R^2	Adjusted R^2	BIC	CV RMSE
2	0.700	0.689	152	5.1
4	0.760	0.743	148	4.6
6	0.780	0.755	153	4.9

For the six-predictor model, residuals show a funnel; $DW = 0.8$ when units are put in their meaningful time order; a Shapiro–Wilk test gives $p = 0.18$; and X_3 has VIF 12.

(a) Interpret each diagnostic and identify the assumption it concerns. Explain which conclusion cannot be justified by the nonsignificant Shapiro–Wilk result. [5 marks]

(b) Explain how VIF is computed, including its auxiliary regression. Calculate the auxiliary R_3^2 implied by VIF 12 and describe the likely effect on coefficient uncertainty. [5 marks]

(c) Select models separately using ordinary R^2, adjusted R^2, BIC and CV RMSE. Explain disagreements and recommend a defensible provisional model. [5 marks]

(d) Compare best subset, forward and backward selection. State how many subsets are considered in an unrestricted best-subset search with six predictors and why a final test set should be protected. [5 marks]

(e) Assume an R data frame `df` contains `y,x1,x2,x3,x4,x5,x6`. Write code to fit the full model, inspect residuals, check VIF, conduct a Durbin–Watson test and obtain best-subset selection criteria. [5 marks]

Question 7 [25 marks]

Ridge, lasso and out-of-sample error

A training/test split is used to compare least squares, ridge and lasso for a continuous outcome. Predictors are numerical. The tuning parameters for ridge and lasso were selected by cross-validation using only the training set. The intercept is not penalized.

Method	Training RMSE	Test RMSE	Nonzero slopes
Least squares	1.8	5.0	12
Ridge	2.4	3.4	12
Lasso	2.6	3.2	5

For a separate four-observation test example, actual outcomes are $(10, 12, 9, 15)$ and predictions are $(11, 10, 10, 13)$.

(a) Write the ridge and lasso penalized least-squares objectives. Explain how their penalties differ and how increasing the tuning parameter affects bias and variance. [5 marks]

(b) Calculate MSE and RMSE for the four-observation example. Explain why the method with the smallest training RMSE is not automatically the preferred predictive model. [5 marks]

(c) Interpret the table and the number of retained predictors. Explain the role of standardizing predictors and why lasso can perform variable selection. [5 marks]

(d) Using built-in `mtcars`, write R code to create a reproducible 70% training split and correctly aligned model matrices for predicting `mpg`. Fit least squares and cross-validated ridge and lasso without using test outcomes to choose the penalties. [5 marks]

(e) Write R code to report train/test RMSE and identify lasso slopes estimated exactly as zero. Explain why repeatedly choosing methods using the same test set would invalidate an unbiased final assessment. [5 marks]

Question 8 [25 marks]

Binary logistic regression: likelihood, odds and prediction

A student-support study models passing ($Y = 1$) versus failing ($Y = 0$), using weekly study hours h and workshop attendance w (1 = attended; 0 = reference). The fitted model and two Wald standard errors are

$$\log \frac{p}{1-p} = -3 + 0.8h - 0.5w, \qquad SE(\widehat{\beta}_h) = 0.25, \quad SE(\widehat{\beta}_w) = 0.40.$$

For code-writing tasks, assume an R data frame `df` contains the numerical columns `pass`, `hours` and `workshop`, with `pass` and `workshop` coded 0/1. No data import is required. Use $z_{0.975} = 1.96$.

(a) Specify the random component, systematic component and link. Write the Bernoulli likelihood for independent observations and explain how maximum likelihood estimates the coefficients. [5 marks]

(b) Interpret the coefficients for hours and attendance in both log-odds and odds-ratio terms, holding the other predictor constant. Explain why a slope is not a fixed probability increase. [5 marks]

(c) Test the hours coefficient using a Wald test and calculate a 95% confidence interval for its odds ratio. Assess the workshop coefficient at 5%. [5 marks]

(d) Calculate the predicted passing probability for a student with $h = 4, w = 1$. Classify that student using thresholds 0.5 and 0.4, and explain the error trade-off when the threshold is lowered. [5 marks]

(e) Write R commands to fit this binary model, extract odds ratios and Wald intervals, and predict a new passing probability. Identify which outcome is the modelled event. [5 marks]

Question 9 [25 marks]

Logistic model comparison, deviance and goodness of fit

Three nested binary logistic models use exactly the same 100 observations:

Model	Estimated parameters	Deviance	Log-likelihood
Null (intercept only)	1	100	−50
Reduced (intercept and x_1)	2	90	−45
Full (adds x_2, x_3)	4	78	−39

The full model has Hosmer–Lemeshow $p = 0.32$. On a distinct test set, a classifier gives the following counts; rows are actual outcomes and columns predicted outcomes:

	Predicted 0	Predicted 1
Actual 0	36	9
Actual 1	6	49

(a) Explain deviance, the saturated model, null deviance and residual deviance. Calculate the reduction from the reduced to the full model. [5 marks]

(b) State the hypotheses for adding x_2, x_3, identify the LRT reference distribution, calculate or specify the p-value and make a decision at 5%. [5 marks]

(c) Compute McFadden's pseudo-R^2 for the full model relative to the null. Explain why it is not the fraction of response variance explained and why ordinary least-squares R^2 should not be substituted. [5 marks]

(d) Interpret the Hosmer–Lemeshow result. Calculate accuracy, sensitivity and specificity from the test matrix and explain why a calibration test and classification accuracy answer different questions. [5 marks]

(e) Assume `df` contains `y,x1,x2,x3`. Write R code to fit all models on a common complete-case sample, compare nested models, compute McFadden's measure and perform the Hosmer–Lemeshow test. [5 marks]

Question 10 [25 marks]

Bayes theorem and categorical Naive Bayes

A club records ten matches. Predictors *Weather* and *Wind* are categorical, and the response is *Win*.

Match	Weather	Wind	Win
1	Rain	Yes	Yes
2	Sun	No	Yes
3	Rain	Yes	Yes
4	Cloud	No	Yes
5	Sun	Yes	Yes
6	Rain	No	Yes
7	Sun	No	No
8	Cloud	No	No
9	Rain	No	No
10	Cloud	No	No

The possible weather levels are Rain, Sun and Cloud; wind levels are Yes and No. Predict the outcome when Weather=Rain and Wind=Yes. Use empirical, unsmoothed class priors; apply add-one smoothing to **every** categorical conditional probability for the smoothed calculation.

(a) State Bayes' theorem and distinguish prior, likelihood and posterior. Calculate both class priors from the table. [5 marks]

(b) Calculate the unsmoothed class-conditional probabilities and both unnormalized posterior scores. Explain the zero-frequency problem in this example. [5 marks]

(c) Recalculate both posterior scores with the specified smoothing, normalize them to posterior probabilities and give the predicted class. [5 marks]

(d) Explain the conditional-independence assumption. Discuss whether a strong marginal association between Weather and Wind alone would invalidate that assumption. [5 marks]

(e) Write R code to enter the data as factors, fit a smoothed Naive Bayes classifier and obtain both the predicted class and posterior probabilities. Explain why evaluating only on these training observations can overstate future accuracy. [5 marks]

Question 11 [25 marks]

LDA, QDA and held-out classification

An analyst uses the four continuous measurements in built-in `iris` to predict the three-level `Species` response. The following two illustrative test matrices refer to the same 30 flowers. Rows are actual classes and columns predicted classes, in order setosa, versicolor, virginica.

LDA	S	Ve	Vi	QDA	S	Ve	Vi
S	10	0	0	S	10	0	0
Ve	0	8	2	Ve	0	9	1
Vi	0	1	9	Vi	0	2	8

(a) State the class-conditional distributional assumptions for LDA and QDA. Explain how they use class priors and Bayes' theorem to classify a new flower. [5 marks]

(b) Explain why LDA produces linear boundaries and QDA generally produces quadratic boundaries. Compare how many covariance parameters they require for four predictors and three classes. [5 marks]

(c) Calculate both accuracies and the recall for versicolor for each model. State what the comparison does and does not establish about which model is better. [5 marks]

(d) Write R code for a reproducible stratified split taking 40 flowers per species for training and the remaining 10 per species for testing. Fit both classifiers on the same training observations. [5 marks]

(e) Write R code to form correctly labelled confusion matrices and calculate test accuracy. Explain why equal test accuracy need not mean identical predictions or equal practical value. [5 marks]

Question 12 [25 marks]

Multinomial logistic regression and reference categories

Transport choice has three unordered outcomes: car, bus and train. Car is the reference. A fitted model uses distance x and has

$$g_{bus}(x) = \log \frac{\pi_{bus}}{\pi_{car}} = -1 + 0.4x, \qquad g_{train}(x) = \log \frac{\pi_{train}}{\pi_{car}} = -0.5 + 0.2x.$$

For the code-writing task, an R data frame `travel` contains `mode` with these three labels and numerical `distance`. No data import is required.

(a) Justify a multinomial model and state the number of baseline-category logit equations for a general K-category response. Explain why imposing a numeric score on these transport modes is inappropriate. [5 marks]

(b) Derive expressions for all three category probabilities. Evaluate them at $x = 5$, check that they sum to one and give the predicted class under a largest-probability rule. [5 marks]

(c) Interpret the distance coefficient for bus versus car as an odds ratio. Explain why it does not necessarily imply that the absolute probability of choosing bus increases. [5 marks]

(d) Rewrite both comparison equations if train becomes the reference category. Explain what changes in the parameters and what happens to the fitted category probabilities. [5 marks]

(e) Write R code to set car as the reference, fit the model, obtain coefficient estimates, predict probabilities at distance 5 and classify that traveller. [5 marks]

Question 13 [25 marks]

Ordinal logistic regression and cumulative probabilities

Satisfaction is ordered Low < Medium < High. A proportional-odds model uses

$$\log \frac{P(Y \le k \mid x)}{P(Y > k \mid x)} = \tau_k - \beta x, \qquad \tau_1 = -1, \quad \tau_2 = 1, \quad \beta = 0.8.$$

For an R application, use the built-in `MASS::housing` data. Its response is `Sat`, predictors are `Infl,Type,Cont`, and `Freq` contains the observed frequency for each combination.

(a) Explain why an ordinal model is appropriate. Define the two cutpoints and the proportional-odds assumption, distinguishing it from a multinomial model with separate slopes. [5 marks]

(b) At $x = 1$, calculate both cumulative probabilities and then each of the three category probabilities. Check their sum and classify by maximum probability. [5 marks]

(c) Interpret $\beta = 0.8$ using higher-versus-lower cumulative odds and lower-versus-higher cumulative odds. Explain why the sign convention must be specified. [5 marks]

(d) Write R code to fit the housing proportional-odds model with an explicitly ordered response, frequency weights and a Hessian. Explain the roles of the weights and Hessian. [5 marks]

(e) Write R code to predict category probabilities for one fully specified housing profile. Explain what a serious violation of proportional odds would mean and why treating Low, Medium and High as equally spaced scores is an additional assumption. [5 marks]

Question 14 [25 marks]

Poisson probabilities and the components of a GLM

A service desk receives an average of three independent arrivals per hour under a homogeneous Poisson process. For a regression study, equal-length observation periods have complaint counts Y and a treatment indicator z (1 = new procedure, 0 = control). The fitted regression is

$$Y_i \mid z_i \sim \text{Poisson}(\mu_i), \qquad \log \mu_i = \log(5) + \log(0.8) z_i.$$

For the R application, use built-in `warpbreaks`, with count response `breaks` and factors `wool,tension`.

(a) Calculate the probability of at least three desk arrivals in the next hour. State the probability formula and show how the complementary event is used. [5 marks]

(b) Identify the random component, systematic component and link for the complaint model. Show how the log link guarantees a positive conditional mean. [5 marks]

(c) Calculate the expected complaint counts for control and treatment, interpret the exponentiated treatment coefficient and give both conditional variances. [5 marks]

(d) An analyst finds variance exceeding the mean after pooling all observation periods. Explain why this alone does not demonstrate conditional overdispersion. State what comparison is relevant to a Poisson regression assumption. [5 marks]

(e) Write R commands for the arrival probability and for fitting the warpbreaks Poisson GLM, extracting multiplicative effects and predicting an expected count for wool A at high tension. [5 marks]

Question 15 [25 marks]

Exponential waiting times, Gamma means and log transformation

Service waiting time T is Exponential with rate $\lambda = 0.2$ per minute. A separate cost study uses a Gamma GLM with log mean

$$\log \mu = \log(100) + 0.2x, \qquad \mathrm{Var}(Y \mid x) = \phi\mu^2, \quad \phi = 0.5.$$

For code applications, use `USArrests` for a Gamma model of `Assault` against `Murder`, `UrbanPop` and `Rape`, and `Puromycin` for the lab-style log-transformed model of `rate` against `conc`. Treat the responses as positive continuous measurements for these exercises.

(a) Calculate $E(T)$ and $P(2 < T < 4)$. Write the Exponential density, distinguishing its rate from its mean, and give R commands for the probability. [5 marks]

(b) For an Exponential regression with $\log E(T \mid x) = a + bx$, express the rate as a function of x. Interpret positive b on both expected waiting time and rate. [5 marks]

(c) For the Gamma cost model, calculate conditional means and variances at $x = 0$ and $x = 1$. Interpret the multiplicative slope effect and contrast its variance function with Poisson regression. [5 marks]

(d) Write R code for the specified Gamma model and for the Puromycin log-transformed linear model. Extract their exponentiated coefficients and explain what each fitted model directly targets. [5 marks]

(e) Explain why `lm(log(rate) ~ conc)` is not automatically an Exponential-distribution regression. Under normal errors on the log scale, distinguish exponentiated fitted log means from original-scale arithmetic means; explain why these models should not be treated as interchangeable. [5 marks]

Question 16 [25 marks]

Bayesian logistic regression and posterior prediction

A binary-response model has $\text{logit}(p_i) = \beta_0 + \beta_1 x_i$, with independent priors $\beta_0 \sim N(0, 10^2)$ and $\beta_1 \sim N(0, 2.5^2)$. The following eight *paired* toy posterior draws illustrate computations; they are too few for reliable Monte Carlo inference.

Draw	1	2	3	4	5	6	7	8
β_0	−1.8	−1.5	−2.0	−1.7	−1.9	−1.6	−2.1	−1.4
β_1	−0.1	0.2	0.4	0.6	0.7	0.9	1.1	1.2

(a) Write the likelihood, joint prior and posterior up to proportionality. Identify the role of the marginal likelihood and contrast maximum likelihood with maximum a posteriori estimation. [5 marks]

(b) Using the toy draws, estimate the posterior mean slope and $P(\beta_1 > 0 \mid D)$. Explain the influence of the slope prior and why posterior probabilities depend on modelling assumptions. [5 marks]

(c) Explain a 95% credible interval for the slope and contrast it with a frequentist confidence interval. Write R code to calculate an equal-tailed credible interval from the slope draws. [5 marks]

(d) At $x_{new} = 2$, write and evaluate the Monte Carlo estimate of the posterior predictive event probability using paired draws. Compare it with the probability obtained by substituting posterior mean coefficients. [5 marks]

(e) Write R code to compute posterior odds-ratio draws, summarize their mean and interval, and calculate the predictive probability. Explain why $E(e^{\beta_1} \mid D)$ generally differs from $e^{E(\beta_1 \mid D)}$ and why draw pairing must be preserved. [5 marks]

Question 17 [25 marks]

Randomized blocks and critical evaluation of evidence

Three fertilizers are randomly assigned to separate plots within each of four soil blocks, once per block. Yields are:

Block	A	B	C
1	20	24	31
2	24	29	34
3	19	23	29
4	23	28	35

For checking arithmetic, total SS is 298.9167, treatment SS is 233.1667 and block SS is 63.5833. A separate nonrandomized university pilot reports that failure fell from 13% to 8% ($p = 0.018$) after introducing a prediction system, but the intervention students had higher prior GPAs.

(a) Explain why an RCBD is suitable and state its additive model, experimental unit and assumptions. Describe a valid treatment randomization within blocks. [5 marks]

(b) Complete the RCBD ANOVA table, including error SS, all degrees, mean squares and both F statistics. State the treatment null hypothesis and interpret a significant block effect. [5 marks]

(c) Explain why fitting a treatment-by-block interaction with one observation per cell does not leave an independently estimable residual error. Contrast the design with a CRD. [5 marks]

(d) Write R code to create the yield data, fit the additive blocked model and inspect its ANOVA. State what the response ordering and factor construction must preserve. [5 marks]

(e) Critically evaluate the university pilot. Define the observed effect size, identify the confounding issue, propose a stronger design and suitable binary-outcome model, and explain why $p = 0.018$ alone does not justify implementation. [5 marks]

Question 18 [25 marks]

Principal component analysis: geometry, scaling and interpretation

Two centered measurements have sample covariance matrix

$$S = \begin{pmatrix} 4 & 3 \\ 3 & 4 \end{pmatrix}.$$

A centered observation is $x = (2, 0)^\top$. For a separate practical exercise, use all four numerical variables in built-in `USArrests`; their units and scales differ.

(a) Explain PCA's objective and the roles of loading vectors and scores. State the unit-length and orthogonality constraints on the first two loading vectors. [5 marks]

(b) Calculate the eigenvalues and normalized eigenvectors of S. Use loading vectors whose first nonzero element is positive, and state the variance explained by each component. [5 marks]

(c) Calculate both PC scores for $x = (2, 0)^\top$ and reconstruct the observation using only PC1. Quantify the squared reconstruction error and explain the information lost. [5 marks]

(d) Write R code to perform centered and standardized PCA on USArrests, extract loadings and scores, and compute the individual and cumulative proportions of variance explained. [5 marks]

(e) Explain how to use a scree plot and cumulative variance in choosing a component count. Explain why a high proportion of predictor variance does not guarantee good response prediction, and why zero correlation of PC scores need not imply independence. [5 marks]

Question 19 [25 marks]

Time-series theory: moving averages and stationarity

Let Z_t be independent identically distributed innovations with mean zero and variance σ^2, and let

$$Y_t = Z_t + 0.5Z_{t-1} - 0.25Z_{t-2}.$$

Separately, consider a random walk $W_t = W_{t-1} + Z_t$ with fixed $W_0 = 0$, and a stationary AR(1) model $X_t = c + 0.6X_{t-1} + Z_t$.

(a) Identify the order of the moving-average process and derive its mean and variance. State whether it is weakly stationary and justify the answer. [5 marks]

(b) Derive $\gamma(1)$ and $\gamma(2)$ for Y_t, and explain why autocovariances vanish for lags larger than 2. Calculate the corresponding autocorrelations. [5 marks]

(c) For the random walk, derive the mean and variance at time t, explain its nonstationarity and write its first-differenced process. Identify its nonseasonal ARIMA order. [5 marks]

(d) Calculate the stationary mean and variance of the AR(1) model. Contrast an autoregressive process, a moving-average stochastic process and a moving-average smoother. [5 marks]

(e) Write R statements to simulate a reproducible length-200 sample from the MA process, plot it in black and inspect its ACF. Explain why an isolated sample ACF spike beyond theoretical lag 2 does not disprove the MA(2) model. [5 marks]

Question 20 [25 marks]

Seasonal time series: differencing, ARIMA and R interpretation

Monthly measurements have annual seasonality. For a worked code exercise use built-in `nottem`, a monthly temperature series. For a separate model-comparison exercise, maximum-likelihood fits to the **same original response series** have the following supplied results, all with seasonal period 12 and the same differencing orders:

Model	Order $(p,d,q)(P,D,Q)_{12}$	AIC	Residual check p
M1	$(1,0,1)(0,1,1)_{12}$	312	0.21
M2	$(2,0,1)(0,1,1)_{12}$	314	0.28
M3	$(0,0,1)(0,1,1)_{12}$	319	0.01

An ADF test on a seasonally differenced series gives $p = 0.02$ under an appropriate test specification. The displayed model-comparison values are illustrative, not promised outputs for nottem.

(a) Distinguish trend, seasonality and cycles. Explain additive versus multiplicative decomposition and give R code to inspect and decompose nottem using an appropriate choice. [5 marks]

(b) Write one seasonal difference at period 12 using observations and the backshift operator. Expand combined ordinary and seasonal differencing, and explain the difference between removing seasonality by differencing and subtracting an estimated seasonal component. [5 marks]

(c) State the ADF null and alternative, interpret $p = 0.02$ and write code to inspect the seasonally differenced series, its ACF/PACF and an ADF test. Explain why a small ADF p-value alone does not verify every stationarity condition. [5 marks]

(d) Select a defensible model using the supplied AIC values and residual checks. Write M1 in backshift notation and identify every order. Explain why the residual check is relevant beyond minimizing AIC. [5 marks]

(e) Write R code to fit all three candidate seasonal models directly to the original nottem series and compare their AIC values. Explain how manually seasonal-differencing first and then specifying $D = 1$ again would change the model. [5 marks]

END OF QUESTIONS

Worked answers and indicative marking guide

Each part carries 5 marks. Award credit for correct methods, stated assumptions, valid calculations and contextual interpretation. Equivalent R code is acceptable. A correct test decision with an incorrect interpretation is not a complete answer. For design questions, well-justified alternatives can receive full credit.

Suggested allocation within a part. Typically 2 marks for the method or code, 2 for a correct calculation or implementation, and 1 for interpretation. Adapt this allocation to the task rather than requiring identical wording. Carry forward reasonable rounding differences.

Important conventions. Wald intervals and profile-likelihood intervals need not match. The solutions label Wald calculations explicitly. Exponential waiting-time models and log-transformed exponential-curve fits use different distributional assumptions. Tests with large p-values provide insufficient evidence against a null; they do not prove assumptions or perfect fit.

Using the source. Set `\showsolutionsfalse` in the preamble to compile a questions-only paper. Each file is self-contained and can be compiled independently with pdfLaTeX. The answer appendix is enabled by default for self-study.

Answer 1: Comparing independent population means

(a) Let $\delta = \mu_A - \mu_B$. Test $H_0 : \delta = 10$ against $H_1 : \delta \neq 10$. Use Welch's two-sample t-test. There is no matching or repeated measurement linking A to B.

(b) $\bar{x}_A = 232.4$, $\bar{x}_B = 217.4$, $s_A^2 = 318.4889$, $s_B^2 = 219.8222$. Thus $\widehat{\delta} = 15$ and

$$SE = \sqrt{318.4889/10 + 219.8222/10} = 7.33697, \qquad t = (15 - 10)/7.33697 = 0.68148.$$

The Welch–Satterthwaite degrees are

$$\nu = \frac{(s_A^2/10 + s_B^2/10)^2}{(s_A^2/10)^2/9 + (s_B^2/10)^2/9} = 17.41495.$$

(c) $p = 2P(t_{17.41495} \geq |0.68148|) = 0.50453$. Do not reject H_0; evidence is insufficient that the population difference differs from 10. This does not establish that it equals 10.

(d) $15 \pm t_{0.975,17.41495}(7.33697) = (-0.4516, 30.4516)$. The interval contains 10, agreeing with the nonrejection. Whether it contains zero answers a different hypothesis.

(e) A complete R solution is:

```
A <- c(195,225,265,231,225,236,245,235,241,226)
B <- c(224,192,197,211,208,235,229,231,228,219)
c(mean(A), mean(B), var(A), var(B))
res <- t.test(A, B, mu = 10, alternative = "two.sided",
              var.equal = FALSE, conf.level = 0.95)
res
res$conf.int
# If equal variances are substantively justified:
t.test(A, B, mu = 10, var.equal = TRUE)
```

The pooled method uses $s_p^2 = [9s_A^2 + 9s_B^2]/18$ and 18 degrees. Equality of sample sizes alone does not justify equal variances.

Answer 2: Paired measurements and the direction of a claim

(a) Define $D = \text{After} - \text{Before}$. At the null boundary test $H_0 : \mu_D = 2$ against $H_1 : \mu_D > 2$ (the full null is $\mu_D \leq 2$). Each pair belongs to the same student, so the scores within it are dependent.

(b) The differences are $(5, 3, 4, 2, 6, 4, 5, 3)$. Their mean is 4 and $s_D = 1.30931$, so $SE = 1.30931/\sqrt{8} = 0.46291$. The statistic is $t = (4 - 2)/0.46291 = 4.32049$, with 7 degrees under the boundary null.

(c) $p = P(t_7 \geq 4.32049) = 0.001739$. Reject H_0; there is evidence that mean improvement exceeds 2 marks. The two-sided interval is $4 \pm t_{0.975,7}(0.46291) = (2.9054, 5.0946)$ marks.

(d) For example:

```
before <- c(52,60,55,48,70,65,58,62)
after <- c(57,63,59,50,76,69,63,65)
d <- after - before
c(mean(d), sd(d))
t.test(after, before, paired = TRUE, mu = 2,
       alternative = "greater")
t.test(d, mu = 2, conf.level = 0.95)$conf.int
qqnorm(d); qqline(d)
shapiro.test(d)
```

The paired normality assumption concerns d, not the two score distributions separately. A nonsignificant Shapiro test is not proof of normality.

(e) Practice effects, other teaching, or maturation could explain before–after changes. Randomize comparable students to workshop and control groups, measure both groups before and after, and compare improvements. Random assignment strengthens a causal claim; paired change alone does not isolate the workshop effect.

Answer 3: Proportions, confidence intervals and variance inference

(a) $H_0 : p_A = p_B$ versus $H_1 : p_A < p_B$. The sample proportions are 0.12 and 0.18, and $\hat{p} = (24 + 45)/(200 + 250) = 0.153333$. The pooled $SE = \sqrt{\hat{p}(1-\hat{p})(1/200 + 1/250)} = 0.034182$. Thus $z = -0.06/SE = -1.75532$ and $p = 0.039603$. Reject at 5% in favour of lower defect probability for A.

(b) The unpooled $SE = \sqrt{0.12(0.88)/200 + 0.18(0.82)/250} = 0.033442$. The interval is $-0.06 \pm 1.96SE = (-0.12555, 0.00555)$. A two-sided 95% interval corresponds to a two-sided 5% test, not the one-sided test used in (a); the Wald interval and pooled score test also use different standard errors.

(c) Use:

```
prop.test(c(24,45), c(200,250),
          alternative = "less", correct = FALSE)
ph <- c(24/200, 45/250)
se <- sqrt(ph[1]*(1-ph[1])/200 + ph[2]*(1-ph[2])/250)
diffhat <- ph[1] - ph[2]
diffhat + c(-1,1)*qnorm(0.975)*se
```

For two groups with no continuity correction, the reported chi-square statistic equals the pooled signed z statistic squared. Its sign is not reported, but R uses the alternative's direction for the one-sided p-value.

(d) Test $H_0 : \sigma^2 = 4$ against $H_1 : \sigma^2 > 4$. The statistic is $19(9)/4 = 42.75$, with 19 degrees, giving $p = 0.0014023$. Reject; there is evidence of variance above 4.

(e) An implementation is:

```
n <- 20; s2 <- 9; sigma02 <- 4
stat <- (n-1)*s2/sigma02
pchisq(stat, df = n-1, lower.tail = FALSE)
(n-1)*s2 / qchisq(c(0.975,0.025), df = n-1)
```

The variance interval is approximately $(5.2051, 19.1994)$ units squared. Exact chi-square inference assumes a random sample from a normal population. A standard deviation must be squared before entering a variance formula.

Answer 4: Completely randomized design, ANOVA and Fisher LSD

(a) A plot is an experimental unit; fertilizer is the factor with levels A, B, C. Yield is the response and there are four plot replications per level. The model is $Y_{ij} = \mu + \tau_i + \varepsilon_{ij}$ with independent normal, zero-mean errors of common variance. Randomization and comparable plots support the design.

(b) $H_0 : \mu_A = \mu_B = \mu_C$; the alternative is that not all means are equal. Means are $(19.5, 23.5, 27.5)$ and the grand mean is 23.5. $SS_T = 4[(19.5 - 23.5)^2 + 0 + (27.5 - 23.5)^2] = 128$. Each group has within-group SS 5, so $SS_E = 15$ and $SS_{Total} = 143$.

(c) The table is:

Source	SS	df	MS	F
Treatment	128	2	64	38.4
Error	15	9	1.6667	
Total	143	11		

Since $38.4 > 4.2565$, reject equality. The p-value is approximately 0.00003921; at least one population mean differs.

(d) $LSD = 2.2622\sqrt{(15/9)(1/4 + 1/4)} = 2.0651$. The absolute pair differences are 4, 8 and 4, so all exceed LSD. The omnibus test alone does not identify the pairs. Unadjusted LSD controls individual comparison errors but not generally the familywise error across many comparisons.

(e) For example:

```
df <- data.frame(
  fertilizer = factor(rep(c("A","B","C"), each = 4)),
  yield = c(18,20,19,21,22,24,23,25,26,28,27,29))
fit <- aov(yield ~ fertilizer, data = df)
summary(fit)
qqnorm(residuals(fit)); qqline(residuals(fit))
shapiro.test(residuals(fit))
car::leveneTest(yield ~ fertilizer, data = df)
agricolae::LSD.test(fit, "fertilizer", p.adj = "none")
```

A large Levene p-value means insufficient evidence of unequal variances; a small sample may give low power to detect a real difference.

Answer 5: Regression in matrix form and multiple responses

(a) $X = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \end{pmatrix}$ is 5×2 and $Y = \begin{pmatrix} 2 & 1 \\ 3 & 3 \\ 7 & 2 \\ 7 & 5 \\ 11 & 4 \end{pmatrix}$ is 5×2. Two responses make the analysis multivariate. Multiple regression instead refers to several predictor variables for a response.

(b) $X^\top Y = \begin{pmatrix} 30 & 15 \\ 82 & 38 \end{pmatrix}$ and $\widehat{B} = \begin{pmatrix} 1.6 & 1.4 \\ 2.2 & 0.8 \end{pmatrix}$. Hence $\widehat{y}_1 = 1.6 + 2.2x$ and $\widehat{y}_2 = 1.4 + 0.8x$.

(c) $\widehat{y}_1 = (1.6, 3.8, 6.0, 8.2, 10.4)^\top$ and $e_1 = (0.4, -0.8, 1.0, -1.2, 0.6)^\top$. Thus $SSE = 3.6$. Since $\bar{y}_1 = 6$, $TSS = 52$, so $R^2 = 1 - 3.6/52 = 0.93077$. About 93.08% of the sample variation about the response mean is explained by the fitted linear relationship; this does not demonstrate causation.

(d) Suitable code is:

```
df <- data.frame(x = 0:4, y1 = c(2,3,7,7,11),
                 y2 = c(1,3,2,5,4))
f1 <- lm(y1 ~ x, data = df)
f2 <- lm(y2 ~ x, data = df)
fm <- lm(cbind(y1,y2) ~ x, data = df)
coef(fm)
cor.test(df$x, df$y1, method = "pearson")
predict(fm, newdata = data.frame(x = 2.5))
```

The two predictions are 7.1 and 3.4. A conventional correlation t-test uses a normal/bivariate-normal sampling model; association does not establish a causal effect.

(e) Add three indicators for four levels, with one level omitted as reference. The design has columns for intercept, x and three indicators, so dimensions are 5×5. All four indicators sum to the intercept and cause rank deficiency. Even if a particular 5×5 design were full rank, it would leave zero error degrees: substantially more observations are needed for useful inference.

Answer 6: Diagnostics and selection of a multiple regression model

(a) The funnel suggests heteroscedasticity; $DW = 0.8$ suggests positive first-order residual autocorrelation in the meaningful ordering. Shapiro $p = 0.18$ gives no strong evidence against normality, but does not establish normality. VIF 12 indicates substantial linear dependence of X_3 on other predictors. Independence would also need reconsideration despite the initial working assumption.

(b) Regress X_3 on the other five predictors, including an intercept. $12 = 1/(1 - R_3^2)$ gives $R_3^2 = 11/12 = 0.91667$. Collinearity inflates the coefficient variance and can produce unstable estimates and large standard errors. It does not by itself prove a predictor has no predictive value.

(c) Ordinary and adjusted R^2 both choose six; BIC and CV RMSE choose four. Training fit can reward complexity, whereas BIC penalizes it and CV estimates performance on omitted data. Provisionally prefer four for prediction and parsimony, then assess its diagnostics and relevant subject knowledge. A criterion does not repair violated assumptions.

(d) Best subset considers every subset, $2^6 = 64$ including the null. Forward selection starts at the intercept-only model and adds predictors; backward starts at the full model and removes predictors. Greedy paths need not find the best subset. If data have temporal dependence, random-fold CV can also mislead; use appropriately ordered validation. Reserve a final test set until selection is complete.

(e) Example commands are:

```
full <- lm(y ~ x1+x2+x3+x4+x5+x6, data = df)
plot(fitted(full), residuals(full)); abline(h = 0)
qqnorm(residuals(full)); qqline(residuals(full))
shapiro.test(residuals(full))
car::vif(full)
# df must already be in the meaningful time order:
lmtest::dwtest(full)
bs <- leaps::regsubsets(y ~ x1+x2+x3+x4+x5+x6,
                        data = df, nvmax = 6)
ss <- summary(bs)
cbind(RSS = ss$rss, R2 = ss$rsq, Cp = ss$cp,
      BIC = ss$bic, AdjR2 = ss$adjr2)
which.min(ss$bic)
which.max(ss$adjr2)
```

Answer 7: Ridge, lasso and out-of-sample error

(a) With unpenalized β_0, ridge minimizes $\sum_i (y_i - \beta_0 - x_i^\top \beta)^2 + \lambda \sum_j \beta_j^2$; lasso replaces the penalty by $\lambda \sum_j |\beta_j|$. Increasing λ shrinks coefficients, usually reducing variance while increasing bias; excessive shrinkage underfits.

(b) Errors have squares $(1, 4, 1, 4)$; $MSE = 10/4 = 2.5$ and $RMSE = \sqrt{2.5} = 1.58114$. Training error can be small because a model fits noise; unseen-data performance is the target.

(c) The least-squares train/test gap suggests overfitting. Both penalized fits improve this test result; lasso has the smallest listed test RMSE and five retained slopes. This one split does not prove universal superiority. Standardization prevents units from determining penalty strength. The L_1 penalty can yield exact zero slopes; ridge generally does not.

(d) A reproducible implementation is:

```
set.seed(3081)
df <- mtcars
id <- sample(seq_len(nrow(df)), floor(0.7*nrow(df)))
tr <- df[id, ]; te <- df[-id, ]
x <- model.matrix(mpg ~ ., df)[, -1, drop = FALSE]
xtr <- x[id, , drop = FALSE]; xte <- x[-id, , drop = FALSE]
ytr <- tr$mpg; yte <- te$mpg
ols <- lm(mpg ~ ., data = tr)
foldid <- sample(rep(1:5, length.out = length(ytr)))
ridge <- glmnet::cv.glmnet(xtr, ytr, alpha = 0,
                           foldid = foldid, standardize = TRUE)
lasso <- glmnet::cv.glmnet(xtr, ytr, alpha = 1,
                           foldid = foldid, standardize = TRUE)
```

(e) Evaluate and inspect coefficients as follows:

```
rmse <- function(actual, predicted) {
  sqrt(mean((actual - as.numeric(predicted))^2))
}
rbind(
  OLS = c(train = rmse(ytr, predict(ols, tr)),
          test = rmse(yte, predict(ols, te))),
  Ridge = c(train = rmse(ytr, predict(ridge, xtr, s="lambda.min")),
            test = rmse(yte, predict(ridge, xte, s="lambda.min"))),
  Lasso = c(train = rmse(ytr, predict(lasso, xtr, s="lambda.min")),
            test = rmse(yte, predict(lasso, xte, s="lambda.min"))))
b <- as.matrix(coef(lasso, s = "lambda.min"))
rownames(b)[b[,1] == 0 & rownames(b) != "(Intercept)"]
```

These built-in-data results need not match the illustrative table. If the test set repeatedly guides method selection, its error becomes optimistic; retain a fresh assessment set or use a nested evaluation design.

Answer 8: Binary logistic regression: likelihood, odds and prediction

(a) $Y_i \mid x_i \sim \text{Bernoulli}(p_i)$, $\eta_i = \beta_0 + \beta_h h_i + \beta_w w_i$, and $g(p_i) = \log[p_i/(1-p_i)]$. The likelihood is $L(\beta) = \prod_i p_i^{y_i}(1-p_i)^{1-y_i}$, with $p_i = \{1 + e^{-\eta_i}\}^{-1}$. MLE chooses coefficients maximizing this likelihood (or its logarithm).

(b) An extra study hour increases log-odds by 0.8 and multiplies passing odds by $e^{0.8} = 2.2255$, about a 122.55% increase. Attendance changes log-odds by -0.5 and multiplies odds by $e^{-0.5} = 0.6065$, about a 39.35% decrease relative to nonattendance. These adjusted associations are not causal conclusions. The probability change depends on the initial probability.

(c) Hours: $H_0 : \beta_h = 0$, $H_1 : \beta_h \neq 0$; $z = 0.8/0.25 = 3.2$, two-sided $p \approx 0.00137$. Reject. The coefficient interval is $(0.31, 1.29)$ and the OR interval is $(e^{0.31}, e^{1.29}) = (1.3634, 3.6328)$. Attendance has $z = -1.25$, $p \approx 0.2113$; evidence is insufficient that its adjusted coefficient differs from zero.

(d) $\eta = -3 + 0.8(4) - 0.5 = -0.3$, so $p = 1/(1 + e^{0.3}) = 0.42556$. Classify as fail at 0.5 and pass at 0.4. Lowering a passing threshold can identify more actual passes but can also falsely classify more actual failures as passes.

(e) For the defined study data:

```
fit <- glm(pass ~ hours + workshop, data = df,
           family = binomial(link = "logit"))
summary(fit)
exp(coef(fit))
b <- coef(fit)
se <- sqrt(diag(vcov(fit)))
exp(cbind(lower = b-1.96*se, upper = b+1.96*se))
predict(fit, newdata = data.frame(hours=4, workshop=1),
        type = "response")
# Reproduce the supplied prediction directly:
plogis(-3 + 0.8*4 - 0.5)
```

With numerical coding 0/1, the event is `pass == 1`. Omitting `type="response"` returns the linear predictor, not a probability. The supplied equation permits the numerical parts without the original study records.

Answer 9: Logistic model comparison, deviance and goodness of fit

(a) Deviance is $2(\ell_{sat} - \ell_{fit})$, measuring likelihood discrepancy from a saturated model that fits the observed responses as fully as possible. Null deviance is for the intercept-only model; residual deviance is for the fitted predictor model. For the comparison, $\Delta D = 90 - 78 = 12$.

(b) Test $H_0 : \beta_2 = \beta_3 = 0$ against at least one being nonzero. Under regularity conditions, $\Delta D \sim \chi^2_2$ approximately. $p = P(\chi^2_2 \geq 12) = 0.002479$; reject. The full model improves likelihood fit over the reduced model. The test does not establish that both added slopes individually differ from zero.

(c) $R^2_{McF} = 1 - \ell_{full}/\ell_{null} = 1 - (-39)/(-50) = 0.22$. The full-to-null order in this ratio matters. This is a likelihood improvement measure, not the OLS sum-of-squares variance decomposition. Different pseudo-R^2 definitions have different scales and interpretations.

(d) $p = 0.32$ gives insufficient evidence of the grouped calibration discrepancy tested by Hosmer–Lemeshow, not proof of perfect fit. $Accuracy = (36 + 49)/100 = 0.85$, $Sensitivity = 49/(49 + 6) = 0.89091$, $Specificity = 36/(36 + 9) = 0.80$. Calibration concerns probability agreement; thresholded accuracy concerns class decisions and depends on threshold and class prevalence.

(e) A common complete-case sample avoids comparing likelihoods from different observations:

```
d <- df[complete.cases(df[,c("y","x1","x2","x3")]), ]
m0 <- glm(y ~ 1, data=d, family=binomial())
m1 <- glm(y ~ x1, data=d, family=binomial())
m2 <- glm(y ~ x1+x2+x3, data=d, family=binomial())
anova(m1, m2, test="Chisq")
pchisq(12, df=2, lower.tail=FALSE)
1 - as.numeric(logLik(m2))/as.numeric(logLik(m0))
ResourceSelection::hoslem.test(d$y, fitted(m2), g=10)
```

Use grouped-fit tests only when their group construction and sample size are adequate; interpret them alongside diagnostics and predictive performance.

Answer 10: Bayes theorem and categorical Naive Bayes

(a) $P(C \mid x) = P(x \mid C)P(C)/P(x)$. Here $P(Yes) = 6/10 = 0.6$ and $P(No) = 4/10 = 0.4$. The prior is before observing the new match predictors; likelihood is predictor probability given the class; posterior is updated class probability.

(b) For Yes, $P(Rain \mid Yes) = 3/6$ and $P(WindYes \mid Yes) = 3/6$, giving score $0.6(3/6)(3/6) = 0.15$. For No, the probabilities are $1/4$ and $0/4$, giving score zero. A single unseen conditional feature level annihilates that class score, even though the true probability need not be zero.

(c) For Yes, smoothed probabilities are $(3+1)/(6+3) = 4/9$ and $(3+1)/(6+2) = 4/8$. For No, they are $(1+1)/(4+3) = 2/7$ and $(0+1)/(4+2) = 1/6$. Scores are $s_Y = 0.6(4/9)(4/8) = 2/15$ and $s_N = 0.4(2/7)(1/6) = 2/105$. Hence $P(Yes \mid x) = 7/8 = 0.875$ and $P(No \mid x) = 1/8 = 0.125$. Predict Yes.

(d) The required factorization is $P(Weather, Wind \mid C) = P(Weather \mid C)P(Wind \mid C)$. Marginal association can arise from mixing classes; it does not alone determine dependence within classes. Check the assumption conditionally, recognizing limited data.

(e) A self-contained implementation is:

```
df <- data.frame(
  Weather = factor(c("Rain","Sun","Rain","Cloud","Sun",
                     "Rain","Sun","Cloud","Rain","Cloud"),
                   levels=c("Rain","Sun","Cloud")),
  Wind = factor(c("Yes","No","Yes","No","Yes",
                  "No","No","No","No","No"),
                levels=c("No","Yes")),
  Win = factor(c(rep("Yes",6),rep("No",4))))
fit <- e1071::naiveBayes(Win ~ Weather+Wind, data=df, laplace=1)
new <- data.frame(
  Weather=factor("Rain", levels=levels(df$Weather)),
  Wind=factor("Yes", levels=levels(df$Wind)))
predict(fit, newdata=new, type="class")
predict(fit, newdata=new, type="raw")
```

The fitted training observations helped estimate the probabilities, so resubstitution accuracy is optimistic. Use sufficiently large held-out data or a suitable validation scheme.

Answer 11: LDA, QDA and held-out classification

(a) Both model $X \mid C = k$ as multivariate normal with class-specific means. LDA assumes one common covariance matrix; QDA estimates a separate covariance per class. Combine a class density with its prior, then normalize to posterior class probabilities and select the largest (for equal misclassification costs).

(b) In LDA, shared quadratic terms cancel in pairwise log-density comparisons, leaving linear boundaries. With unequal covariances, quadratic terms generally remain in QDA. A symmetric 4×4 covariance has $4(5)/2 = 10$ free entries; LDA estimates 10 covariance parameters and QDA estimates $3(10) = 30$, excluding means and priors.

(c) Both accuracies are $27/30 = 0.90$. Versicolor recall is $8/10 = 0.80$ for LDA and $9/10 = 0.90$ for QDA. QDA trades an additional correct versicolor for one fewer correct virginica. One small test split does not prove equal population performance or universal superiority.

(d) A stratified split is:

```
df <- iris
set.seed(2222)
ids <- split(seq_len(nrow(df)), df$Species)
train_id <- unlist(lapply(ids, function(z) sample(z, 40)),
                   use.names=FALSE)
train <- df[train_id, ]; test <- df[-train_id, ]
lda_fit <- MASS::lda(Species ~ ., data=train)
qda_fit <- MASS::qda(Species ~ ., data=train)
```

(e) Evaluate using:

```
p_lda <- predict(lda_fit, newdata=test)$class
p_qda <- predict(qda_fit, newdata=test)$class
cm_lda <- table(Actual=test$Species, Predicted=p_lda)
cm_qda <- table(Actual=test$Species, Predicted=p_qda)
cm_lda; cm_qda
sum(diag(cm_lda))/sum(cm_lda)
sum(diag(cm_qda))/sum(cm_qda)
```

The supplied matrices illustrate interpretation; the reproducible split may give different counts. Error types, class-specific recall and misclassification costs matter even at equal accuracy. QDA's greater flexibility can increase estimation variance in small samples.

Answer 12: Multinomial logistic regression and reference categories

(a) The response has three nominal categories; multinomial logistic regression models them jointly without inventing an order or equal spacing. There are $K-1$ baseline-category equations for K categories.

(b) Let $D = 1 + e^{g_{bus}} + e^{g_{train}}$. Then $\pi_{car} = 1/D$, $\pi_{bus} = e^{g_{bus}}/D$, $\pi_{train} = e^{g_{train}}/D$. At $x = 5$, $g_{bus} = 1$ and $g_{train} = 0.5$, so the probabilities are approximately $(0.18632, 0.50648, 0.30720)$, which sum to 1. Predict bus.

(c) An extra distance unit multiplies π_{bus}/π_{car} by $e^{0.4} = 1.49182$, holding other predictors fixed. In a general multinomial model, normalization also depends on other outcome logits; a relative probability ratio can increase even when an absolute category probability decreases. Direction of an absolute change must be checked using all logits.

(d) With train as reference:

$$\log \frac{\pi_{car}}{\pi_{train}} = 0.5 - 0.2x, \quad \log \frac{\pi_{bus}}{\pi_{train}} = g_{bus} - g_{train} = -0.5 + 0.2x.$$

The coefficient parameterization changes but the same model's fitted probabilities and classifications do not.

(e) R code for the defined study data is:

```
travel$mode <- relevel(factor(travel$mode), ref="car")
fit <- nnet::multinom(mode ~ distance, data=travel, trace=FALSE)
summary(fit)
predict(fit, newdata=data.frame(distance=5), type="probs")
predict(fit, newdata=data.frame(distance=5), type="class")
# Direct calculation from the supplied fitted equations:
eta <- c(car=0, bus=-1+0.4*5, train=-0.5+0.2*5)
exp(eta)/sum(exp(eta))
```

The numerical equations supply everything required for parts (a)–(d); code fitting presumes the defined study data frame exists.

Answer 13: Ordinal logistic regression and cumulative probabilities

(a) Categories have an order but no known equal spacing. τ_1 separates Low from Medium/High; τ_2 separates Low/Medium from High. Proportional odds means the same predictor slope at both cumulative cutpoints; a baseline-category multinomial model generally estimates separate slopes for each nonreference outcome.

(b) At $x = 1$, cumulative logits are -1.8 and 0.2. Thus $F_1 = \text{logit}^{-1}(-1.8) = 0.14185$ and $F_2 = \text{logit}^{-1}(0.2) = 0.54983$. Category probabilities are $(F_1, F_2 - F_1, 1 - F_2) = (0.14185, 0.40798, 0.45017)$. Their sum is 1 and High is most likely.

(c) For a one-unit increase in x, higher-versus-lower cumulative odds multiply by $e^{0.8} = 2.22554$. Lower-versus-higher cumulative odds multiply by $e^{-0.8} = 0.44933$. A positive slope shifts probability towards higher categories under the stated minus-sign parameterization.

(d) Fit the model as follows:

```
data("housing", package="MASS")
housing$Sat <- ordered(housing$Sat,
                       levels=c("Low","Medium","High"))
fit <- MASS::polr(Sat ~ Infl+Type+Cont, weights=Freq,
                  data=housing, method="logistic", Hess=TRUE)
summary(fit)
exp(coef(fit))
```

Frequency weights represent how many observations each row stands for. `Hess=TRUE` requests the Hessian used in covariance/standard-error calculations; it does not test proportional odds.

(e) For example:

```
new <- data.frame(
  Infl=factor("High", levels=levels(housing$Infl)),
  Type=factor("Tower", levels=levels(housing$Type)),
  Cont=factor("High", levels=levels(housing$Cont)))
predict(fit, newdata=new, type="probs")
F <- plogis(c(-1,1)-0.8)
c(Low=F[1], Medium=F[2]-F[1], High=1-F[2])
```

A violation means predictor effects differ across cumulative splits, so one common slope may misrepresent them. A more flexible ordinal or multinomial analysis may be needed. Numeric scores presume equal distance between adjacent categories, which ordering alone does not establish.

Answer 14: Poisson probabilities and the components of a GLM

(a) $P(N = k) = e^{-3}3^k/k!$. Thus $P(N \geq 3) = 1 - P(N \leq 2) = 1 - e^{-3}(1 + 3 + 9/2) = 0.57681$.

(b) The random component is conditional Poisson; systematic component is $\eta_i = \beta_0 + \beta_1 z_i$ with given coefficients; link is $g(\mu) = \log \mu$. The inverse link gives $\mu_i = e^{\eta_i} > 0$. A GLM links the conditional mean, not an observed count or its logarithm; zeros in the count response are allowed.

(c) Control mean $\mu_0 = 5$ and treatment mean $\mu_1 = 5(0.8) = 4$. Treatment is associated with 20% fewer expected complaints for the same exposure. The corresponding conditional variances are 5 and 4 under the basic Poisson model. The exponentiated intercept is a baseline mean, not a treatment ratio.

(d) Mixing observations with distinct means can increase marginal variance: $\text{Var}(Y) = E\{\text{Var}(Y \mid X)\} + \text{Var}\{E(Y \mid X)\} = E(\mu) + \text{Var}(\mu)$. Assess excess variation relative to the fitted conditional mean structure, using appropriate residual/dispersion diagnostics. A pooled variance-mean difference alone is insufficient.

(e) Commands are:

```
1 - ppois(2, lambda=3)
data("warpbreaks")
fit <- glm(breaks ~ wool+tension, data=warpbreaks,
           family=poisson(link="log"))
summary(fit)
exp(coef(fit))
new <- data.frame(
  wool=factor("A", levels=levels(warpbreaks$wool)),
  tension=factor("H", levels=levels(warpbreaks$tension)))
predict(fit, newdata=new, type="response")
sum(residuals(fit, type="pearson")^2)/df.residual(fit)
```

The final ratio is a rough dispersion diagnostic, interpreted with the model specification and sampling design. The warpbreaks example is a separate application and need not reproduce the given complaint equation.

Answer 15: Exponential waiting times, Gamma means and log transformation

(a) $f(t) = \lambda e^{-\lambda t}$ for $t \geq 0$. The rate is 0.2 per minute and $E(T) = 1/\lambda = 5$ minutes. $P(2 < T < 4) = e^{-0.2(2)} - e^{-0.2(4)} = 0.220991$.

```
1/0.2
pexp(4, rate=0.2) - pexp(2, rate=0.2)
```

(b) $\mu(x) = e^{a+bx}$, hence $\lambda(x) = 1/\mu(x) = e^{-(a+bx)}$. One extra unit multiplies mean time by e^b and rate by e^{-b}. Positive b means longer expected waits and smaller rates, under this mean-link convention.

(c) At zero, $\mu_0 = 100$ and $V_0 = 0.5(100)^2 = 5000$. At one, $\mu_1 = 100e^{0.2} = 122.1403$ and $V_1 = 0.5(100e^{0.2})^2 = 7459.1235$. The mean rises by about 22.14%. Gamma has variance proportional to the squared mean; basic Poisson has variance equal to the mean and a count response.

(d) The model commands are:

```
g <- glm(Assault ~ Murder+UrbanPop+Rape,
         data=USArrests, family=Gamma(link="log"))
summary(g)
exp(coef(g))
e <- lm(log(rate) ~ conc, data=Puromycin)
summary(e)
a <- exp(coef(e)[1]); b <- coef(e)[2]
c(a=a, b=b)
```

The Gamma GLM directly models $\log E(Y \mid X)$; exponentiated slopes multiply the conditional arithmetic mean. The linear model directly models $E(\log Y \mid X)$; ae^{bx} exponentiates the fitted log mean.

(e) In general $E(\log Y \mid X) \neq \log E(Y \mid X)$. Log-transforming a response does not specify an Exponential distribution. If $\log Y = \eta + \varepsilon$ with $\varepsilon \sim N(0, \sigma^2)$, then e^{η} is the conditional median and $E(Y \mid X) = e^{\eta + \sigma^2/2}$. The correction illustrates why back-transforming a log-scale fit need not give an unbiased mean prediction. The lab's exponential-curve fit and the lecture's Exponential waiting-time model use different response assumptions.

Answer 16: Bayesian logistic regression and posterior prediction

(a) $L(\beta) = \prod_i p_i^{y_i}(1-p_i)^{1-y_i}$ and the independent normal priors give $p(\beta) = p(\beta_0)p(\beta_1)$. Then $p(\beta \mid D) \propto L(\beta)p(\beta)$; integrating the numerator over coefficients gives the marginal likelihood that normalizes it. MLE maximizes L; MAP maximizes the posterior density and includes the prior.

(b) The toy mean slope is $5/8 = 0.625$; seven of eight slopes exceed zero, giving 0.875. The prior favours moderate slopes around zero, while still allowing positive and negative effects. Combining it with data updates uncertainty. A wider prior conveys greater uncertainty; neither the posterior nor a sign probability is independent of the prior and likelihood.

(c) A 95% credible interval contains 95% posterior probability conditional on the model and data. A frequentist 95% confidence procedure has 95% repeated-sampling coverage, rather than assigning posterior probability to a fixed parameter. Using default R quantiles, the toy equal-tailed interval is approximately $(-0.0475, 1.1825)$; it is only a numerical illustration.

(d) Use $\widehat{P}(Y_{new} = 1 \mid D) = \frac{1}{8}\sum_{m=1}^{8} \text{logit}^{-1}(\beta_0^{(m)} + 2\beta_1^{(m)})$. This is approximately 0.39517. The posterior mean intercept is -1.75, so the plug-in value is $\text{logit}^{-1}(-1.75 + 2(0.625)) = \text{logit}^{-1}(-0.5) = 0.37754$. Averaging nonlinear predictions is generally different from predicting at averaged coefficients.

(e) R calculations are:

```
b0 <- c(-1.8,-1.5,-2.0,-1.7,-1.9,-1.6,-2.1,-1.4)
b1 <- c(-0.1,0.2,0.4,0.6,0.7,0.9,1.1,1.2)
mean(b1)
mean(b1 > 0)
quantile(b1, c(0.025,0.975))
OR <- exp(b1)
c(mean=mean(OR), quantile(OR, c(0.025,0.975)))
mean(plogis(b0 + 2*b1))
plogis(mean(b0) + 2*mean(b1))
```

Exponentiation is nonlinear, so the transformed mean is not generally the mean transformation. Coefficients in a draw come from their joint posterior; independently shuffling columns would change their dependence and predictive distribution. Real MCMC analysis needs many more draws and convergence/mixing checks.

Answer 17: Randomized blocks and critical evaluation of evidence

(a) Soil conditions are a known nuisance source; blocks group comparable plots. Use $Y_{ij} = \mu + \tau_i + b_j + \varepsilon_{ij}$. A plot is an experimental unit. Assume independent, normal, constant-variance errors and additive treatment/block effects. Within each block independently randomize one plot to each of A, B and C.

(b) $SS_E = 298.9167 - 233.1667 - 63.5833 \approx 2.1667$. Using unrounded values:

Source	SS	df	MS	F
Treatment	233.1667	2	116.5833	322.8462
Block	63.5833	3	21.1944	58.6923
Error	2.1667	6	0.3611	
Total	298.9167	11		

$H_0 : \mu_A = \mu_B = \mu_C$ after accounting for blocks. Both F values are highly significant at 5%. A block effect indicates substantial nuisance variation between soil blocks; accounting for it can improve treatment precision.

(c) A full interaction allocates $(3-1)(4-1) = 6$ additional degrees, using all six residual degrees and leaving zero for error. The usual single-observation RCBD therefore assumes additivity. A CRD omits the block term and incorporates block variation into its error.

(d) Construct values row by row by block:

```
df <- data.frame(
  block=factor(rep(1:4, each=3)),
  treatment=factor(rep(c("A","B","C"), times=4)),
  yield=c(20,24,31,24,29,34,19,23,29,23,28,35))
fit <- aov(yield ~ treatment+block, data=df)
summary(fit)
plot(fitted(fit), residuals(fit)); abline(h=0)
qqnorm(residuals(fit)); qqline(residuals(fit))
```

Every yield must remain associated with its correct block and treatment. For these balanced data, additive treatment and block sums of squares are orthogonal.

(e) Failure fell by 5 percentage points, a relative reduction of $5/13 \approx 38.46\%$. Higher prior GPA could explain some or all of the reduction. Prefer a randomized intervention-versus-control comparison, possibly blocking on prior GPA. Model failure as binary with treatment, baseline GPA and relevant pre-intervention predictors; avoid adjusting for outcomes caused by treatment when estimating its total effect. Examine adjusted effect sizes, uncertainty, comparable outcome definitions, costs and generalizability. A prediction system's accuracy is also distinct from proof that the resulting intervention reduces failure.

Answer 18: Principal component analysis: geometry, scaling and interpretation

(a) PC1 maximizes the variance of the centered linear combination $z_1 = \phi_1^\top x$ subject to $\|\phi_1\| = 1$. PC2 maximizes remaining variance subject to unit length and $\phi_2^\top \phi_1 = 0$. Loadings define directions; scores are observations projected along them. PCA does not use a response variable when constructing these directions.

(b) $\det(S - \lambda I) = (4 - \lambda)^2 - 9 = 0$ gives eigenvalues 7 and 1. Choose $\phi_1 = (1, 1)^\top / \sqrt{2}$ and $\phi_2 = (1, -1)^\top / \sqrt{2}$. Their variances are 7 and 1, explaining $7/8 = 87.5\%$ and $1/8 = 12.5\%$. Either eigenvector's negative is equally valid if scores are sign-flipped consistently.

(c) $z_1 = \sqrt{2}$ and $z_2 = \sqrt{2}$. Retaining only PC1 reconstructs $\widehat{x} = z_1 \phi_1 = (1, 1)^\top$. The residual is $(1, -1)^\top$ with squared error 2, equal to z_2^2. PC2's difference-between-variables direction is discarded. For original uncentered data the training means must also be added back.

(d) Use:

```
pca <- prcomp(USArrests, center=TRUE, scale.=TRUE)
pca$rotation      # loading columns
pca$x             # observation score rows
eig <- pca$sdev^2
pve <- eig/sum(eig)
cbind(PVE=pve, Cumulative=cumsum(pve))
plot(seq_along(pve), pve, type="b", col="black",
     xlab="Component", ylab="Proportion of variance")
biplot(pca, col=c("black","black"))
```

Standardization makes each variable's initial variance one, preventing numerical units from dominating the exercise. It changes the analysis relative to PCA of an unscaled covariance matrix.

(e) Seek a parsimonious component count supported by the scree elbow and a context-appropriate cumulative variance target; neither supplies a universal cutoff. Low-variance directions can still predict a response, since unsupervised PCA ignores it. Component scores are uncorrelated by construction; independence is a stronger distributional property, not generally implied by zero covariance.

Answer 19: Time-series theory: moving averages and stationarity

(a) This is MA(2), equivalently ARIMA$(0, 0, 2)$. Its mean is zero. By innovation independence,

$$\gamma(0) = \mathrm{Var}(Y_t) = \sigma^2(1 + 0.5^2 + (-0.25)^2) = 1.3125\sigma^2.$$

Constant coefficients and iid finite-variance innovations yield constant mean, variance, and autocovariance depending only on lag; it is weakly stationary.

(b) Overlapping innovations give $\gamma(1) = \sigma^2[1(0.5) + 0.5(-0.25)] = 0.375\sigma^2$ and $\gamma(2) = -0.25\sigma^2$. Thus $\rho(1) = 0.375/1.3125 = 2/7$ and $\rho(2) = -0.25/1.3125 = -4/21$. There are no shared innovations for larger lags, so the covariance is zero.

(c) $W_t = \sum_{j=1}^{t} Z_j$, so $E(W_t) = 0$ but $\mathrm{Var}(W_t) = t\sigma^2$, which grows with time. It is nonstationary. $(1 - B)W_t = Z_t$ is white noise; the original series is ARIMA$(0, 1, 0)$ without drift.

(d) AR(1) has stationary mean $\mu = c/(1 - 0.6) = 2.5c$ and variance $\sigma^2/(1 - 0.6^2) = \sigma^2/0.64$. An AR uses lagged response values; a stochastic MA uses current and lagged innovations. A moving-average smoother averages nearby observations to estimate a trend and is a different operation.

(e) For example:

```
set.seed(3081)
y <- arima.sim(model=list(ma=c(0.5,-0.25)), n=200, sd=1)
plot(y, col="black", ylab="Y", xlab="Time")
acf(y, lag.max=20, col="black")
```

The theoretical ACF is zero beyond lag 2, but sample estimates fluctuate. Approximate pointwise 95% white-noise limits are about $\pm 1.96/\sqrt{200} = \pm 0.1386$; examining many lags makes occasional exceedances unsurprising. One spike is not a definitive model rejection.

Answer 20: Seasonal time series: differencing, ARIMA and R interpretation

(a) Trend is long-term level change; seasonality repeats at a fixed known period; cycles have no fixed duration. Additive decomposition $y_t = T_t + S_t + R_t$ suits roughly constant seasonal amplitude; multiplicative $y_t = T_t S_t R_t$ suits amplitude proportional to a positive level. Monthly temperatures are naturally treated additively; a Celsius zero is arbitrary, so multiplicative comparisons are inappropriate without substantive justification.

```
data("nottem")
frequency(nottem)
plot(nottem, col="black")
dc <- decompose(nottem, type="additive")
plot(dc, col="black")
```

(b) A seasonal difference is $w_t = y_t - y_{t-12} = (1 - B^{12})y_t$. Combining it with one ordinary difference gives $(1 - B)(1 - B^{12})y_t = y_t - y_{t-1} - y_{t-12} + y_{t-13}$. Differencing compares observations across seasons; subtracting an estimated seasonal component produces a seasonally adjusted level series. They are distinct transformations.

(c) ADF tests a unit-root null against stationarity under its chosen deterministic-term/lag specification. At 5%, $p = 0.02$ rejects the unit-root null. It does not establish stable variance, rule out every seasonal unit root, or validate the entire model.

```
w <- diff(nottem, lag=12, differences=1)
plot(w, col="black")
acf(w, lag.max=48, col="black")
pacf(w, lag.max=48, col="black")
tseries::adf.test(w)
```

(d) M1 has the smallest supplied AIC and no strong evidence of residual autocorrelation from the stated check, so is the provisional choice. M2 is close ($\Delta AIC = 2$), while M3 is worse and has evidence of residual dependence. For R's plus-sign MA convention and no drift:

$$(1 - \phi_1 B)(1 - B^{12})y_t = (1 + \theta_1 B)(1 + \Theta_1 B^{12})\varepsilon_t.$$

Here $(p, d, q) = (1, 0, 1)$ and $(P, D, Q) = (0, 1, 1)$ at period 12. Residual dependence indicates remaining temporal structure even if a model's AIC is relatively small. A large check p-value does not prove white noise.

(e) Fit the original series without applying a manual seasonal difference:

```
m1 <- arima(nottem, order=c(1,0,1),
  seasonal=list(order=c(0,1,1), period=12), method="ML")
m2 <- arima(nottem, order=c(2,0,1),
  seasonal=list(order=c(0,1,1), period=12), method="ML")
m3 <- arima(nottem, order=c(0,0,1),
  seasonal=list(order=c(0,1,1), period=12), method="ML")
AIC(m1,m2,m3)
acf(residuals(m1), col="black")
Box.test(residuals(m1), lag=24, type="Ljung-Box", fitdf=3)
```

The estimated AR/MA coefficient count for M1 is $1 + 1 + 1 = 3$ for the indicated residual-test adjustment. Applying `diff(...,lag=12)` first and then retaining $D = 1$ would apply $(1 - B^{12})$ twice relative to the original series. If fitting the already-differenced series, specify $D = 0$ instead and track the original model order carefully.